



Octopus arms make decisions

A new model was created to represent information flow between Octopus' suckers. <https://digit.in/july19-40>



Quantum tech and medicine

Quantum technology is being used to improve medical imaging to detect diseases faster. <https://digit.in/july19-41>

REINVENTING THE WHEEL

The traditional wheels are simply not good enough for emerging applications.



Credit: DARPA

Aditya Madanapalle |
aditya@digit.in

T

he wheel is a simple yet versatile machine that can be used for a number of purposes.

The most obvious use is to transport things, including people, livestock and goods. The wheel can also be used in a vertical orientation to make pots, used as a steering instrument for ships or cars, and used to generate or transmit energy. The earliest wheels were made around 4,500 BC and were simply wooden or

stone disks with holes in the middle. The earliest applications were for pottery and agricultural carts. Over time, spokes were added to make the wheels lighter, and improve their load bearing performance. While this ancient invention may be taken for granted today, scientists are still working on ways to improve the wheel. These developments are funded by the emerging applications in defence and space exploration.

SHAPE SHIFTING WHEELS

The traditional approach for armoured vehicles has been to slap on more armour. This has resulted in increasing-

ly massive vehicles. At the same time, the capability of enemy projectiles to penetrate the armour has also been increasing. The increase has been asymmetric, with munitions now available that far exceed the capability of armour from protecting the occupants. The US Defense Advanced Research Projects Agency (DARPA) funded research programs in a number of US institutions to develop armoured combat vehicles that used different approaches to get the same results. DARPA wanted armoured vehicles that were faster, more maneuverable, and yet improved the safety and survivability, without simply using